- Blending problems
  - Entities
    - Set of ingredients or raw materials
      - I = \{i\}
    - Set of products
      - J = \{j\}
  - Parameters
    - Blend ratio requirements
      - Lower bound
        - a_{i,j}, \forall i \in I, \forall j \in J
      - Upper bound
        - b_{i,j}, \forall i \in I, \forall j \in J
    - Costs
      - c_i, \forall i \in I
    - Ingredient supply limit
      - s_i
    - Demand of ingredients
      - d_j, \forall j \in J
    - Quality of products
      - Quality score of Ingredient i in Product j
        - q_{i,j}, \forall i \in I, \forall j \in J
      - Quality threshold of Product j
        - Upper limit
          - Q^U_j, \forall j \in J
        - Lower limit
          - Q^L_j, \forall j \in J
    - Production capacity limit
      - Capacity limit of production
        - P^U_j, \forall j \in J
      - Minimum production
        - P^L_j, \forall j \in J
  - Decision Variables
    - Amount of Ingredient i in Product J
      - x_{i,j} \geq 0, \forall i \in I, j \in J
    - To produce j or not
      - y_j \in \{0,1\}, \forall i \in I
  - Objective
    - Minimise total cost of ingredients used
      - \min \sum_{i \in I} \sum_{j \in J} c_i x_{i,j}
    - Maximise overall product quality
      - \max \sum_{i \in I} \sum_{j \in J} q_{i,j} x_{i,j}
  - Constraints
    - Explicit constraints
      - Blend ratios
        - The ratio of Ingredient i in Product j must be at least
          - x_{i,j} \geq a_{i,j} \sum_i x_{i,j}, \forall j \in J
        - The ratio of Ingredient i in Product j must be no more than
          - x_{i,j} \leq b_{i,j} \sum_i x_{i,j}, \forall j \in J
      - Demand requirements
        - Product j produced must be at least meeting the demand
          - \sum_{i \in I} x_{i,j} \geq d_j, \forall j \in J
        - Product j produced must be no more than the demand
          - \sum_{i \in I} x_{i,j} \leq d_j, \forall j \in J
      - Supply limits
        - \sum_{j \in J} x_{i,j} \leq s_i, \forall i \in I
      - Production capacity limits
        - \sum_{i \in I} x_{i,j} \leq U_j, \forall j \in J
      - Quality threshold constraints
        - Total quality score of Product j must be at least the lower threshold
          - \sum_{i \in I} q_{i,j} x_{i,j} \geq Q^L_j, \forall j \in J
        - Total quality score of Product j must be at least the lower threshold
          - \sum_{i \in I} q_{i,j} x_{i,j} \leq Q^U_j, \forall j \in J
    - Implicit constraints
      - Link constraints
        - x_{i,j} \leq P^U_j y_j, \forall i \in I, \forall j \in J
        - x_{i,j} \geq P^L_j y_j, \forall i \in I, \forall j \in J